

Practice Quiz Nine: Phasors: Phasor Math (Easy)

Part One: Given the time-domain expression of a sinusoidal voltage source below, find the phasor-domain representation V_s .

$$v_s(t) = 50\sqrt{2} \sin(800t + 20^\circ) \text{ V}$$

Finally, plot V_s on the complex plane.

Part Two: Given the two voltage phasors below, perform the following operations:

a) $V_1 V_2$

b) $\frac{V_1}{V_2}$

$$V_1 = 3 + 3j$$

$$V_2 = 6 - j$$

Finally, write the answers in polar form.

Suggested time limit: 10 minutes